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ELECTRIC POWER STORAGE DEVICE

Abstract

An electric power storage device includes a plurality of electric power storage units and a housing case. The electric power storage units are arranged in an X direction and laminated in the X direction. Lower surfaces of the electric power storage units are bonded to a lower case with an adhesive member. A protruding portion extending in the X direction is formed in an upper case. The protruding portion protrudes toward the lower case and abuts against an upper surface of the electric power storage units. As a result, the electric power storage units receive a load on the adhesive member side (lower case side) and are pressed down.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031979 filed on Mar. 4, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to an electric power storage device.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2023-46013 (JP 2023-46013 A) discloses a battery pack in which a plurality of battery cells and the bottom of a case are bonded together with a heat transfer member having adhesive property.

SUMMARY

[0004] There is a concern that the adhesion between a battery cell (electric power storage unit) and a case may be released when vibration or the like is applied to a battery pack.

[0005] An object of the present disclosure is to provide a structure in which the adhesion between the electric power storage unit and the case is difficult to be released.

[0006] An electric power storage device according to the present disclosure is an electric power storage device that includes a plurality of electric power storage units, and a housing case including a first case and a second case and housing the electric power storage units in a space defined by the first case and the second case. The electric power storage units are bonded to the first case, the second case has a protruding portion protruding toward the first case, and the protruding portion abuts against the electric power storage units.

[0007] According to this configuration, the electric power storage units are bonded to the first case. The protruding portion of the second case protrudes toward the first case and abuts against the power storage units. The electric power storage units are pressed toward the first case by the protruding portion of the second case, so that it is possible to restrain the adhesion between the electric power storage units and the housing case from being released.

[0008] Preferably, the electric power storage units may be arranged in a lamination direction in the storage case, and the protruding portion may extend in the lamination direction.

[0009] According to this configuration, the electric power storage units can be pressed against the first case side by the protruding portion of the second case.

[0010] Each of the electric power storage units may have a rectangular parallelepiped shape, and may be bonded to the first case at center portions in a longitudinal direction of the electric power storage units, and the protruding portion may abut against the electric power storage units at the center portions in the longitudinal direction of the electric power storage units.

[0011] According to this configuration, the center portion of the electric power storage unit can be bonded to the housing case. Each of the electric power storage units may include a cell assembly in which a plurality of electric power storage cells arranged in a connection direction is electrically connected at connection portions, a cell case that houses the cell assembly, and a reinforcing member that is disposed at the connection portion and reinforces the cell case. In this case, the protruding portion of the second case may abut against the cell case at a position where the reinforcing member is disposed.

[0012] According to this configuration, the protruding portion abuts against the cell case at the position where the reinforcing member is arranged, so that it is possible to suitably press the electric power storage units against the first case.

[0013] Furthermore, the cell case may be bonded to the first case at the position where the reinforcing member is disposed.

[0014] According to this configuration, the cell case is bonded to the first case at the position where the reinforcing member is arranged, so that the force applied to the electric power storage unit by the protruding portion of the second case is suitably transmitted to the bonded location through the reinforcing member.

[0015] According to the present disclosure, it is possible to provide a structure in which the adhesion between the electric power storage unit and the case is difficult to be released.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0017] FIG. 1 is a perspective view that schematically shows an electric power storage device according to an embodiment;

[0018] FIG. 2 is a diagram that schematically shows an example of an electric power storage unit **10**;

[0019] FIG. 3A is a sectional view taken along III-III in FIG. 1;

[0020] FIG. 3B is a sectional view taken along III-III in FIG. 1;

[0021] FIG. 4 is an exploded perspective view of the electric power storage unit according to an embodiment 2;

[0022] FIG. 5 is a sectional view of an electric power storage device according to the embodiment 2; and

[0023] FIG. 6 is a perspective view that schematically shows an electric power storage device **1B** according to a modification.

DETAILED DESCRIPTION OF EMBODIMENTS

[0024] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. The same or corresponding components in the drawings are designated by the same reference signs, and the description thereof will not be repeated. In the drawings, illustrations are not made according to the actual dimensional ratios, and in order to facilitate understanding of structures, the ratios may be changed such that the structures are clarified. The embodiments and modifications described below may be selectively combined as appropriate.

Embodiment 1

[0025] An electric power storage device according to the present embodiment will be described with reference to FIG. 1 to FIG. 3B. FIG. 1 is a perspective view that schematically shows the electric power storage device according to the present embodiment.

[0026] With reference to FIG. 1, the electric power storage device **1** is used, for example, while it is mounted on a vehicle. Examples of the vehicle include a hybrid electric vehicle, a plug-in hybrid electric vehicle, a fuel cell electric vehicle, and a battery electric vehicle. The electric power storage device **1** includes a plurality of electric power storage units **10** and a housing case **20** that houses the electric power storage units **10**.

[0027] The electric power storage unit **10** is a secondary battery, typically a lithium ion battery. The lithium ion battery is a battery that uses lithium as a charge carrier, and it may include not only a general lithium ion secondary battery in which an electrolyte is liquid, but also a so-called all-solid-state battery in which a solid electrolyte is used. Examples of lithium ion batteries include an LFP battery in which lithium iron phosphate is used as a positive electrode active material, and a ternary battery in which nickel manganese cobalt (NMC) is used as a positive electrode active material.

Note that the electric power storage unit **10** is not limited to a lithium ion secondary battery, and it may be composed of a nickel-metal hydride secondary battery or other secondary batteries.

[0028] FIG. **2** is a diagram that schematically shows an example of the electric power storage unit **10**. In the present embodiment, the electric power storage unit **10** is a lamination type lithium ion battery in which a plurality of battery cells is laminated in a cell case, each of the battery cells being formed by laminating a positive electrode sheet (current collector) coated with a positive electrode active material and a negative electrode sheet (current collector) coated with a negative electrode active material with a separator interposed therebetween. Note that the electric power storage unit **10** may be a bipolar lithium ion battery in which a positive electrode active material is coated on one side of a current collector while a negative electrode active material is coated on the other side of the current collector, and these current collectors are laminated with a separator interposed therebetween.

[0029] Referring to FIG. **2**, the electric power storage unit **10** has a rectangular parallelepiped shape, and includes an upper surface **11**, a lower surface **12**, a pair of short side surfaces **13**, **14**, and a pair of long side surfaces **15**, **16** with the long side (longest side) extending in a Y direction. The electric power storage unit **10** further includes a positive electrode terminal **17** and a negative electrode terminal **18**. The positive electrode terminal **17** is provided on one of the pair of short side surfaces **13**, **14**, and the negative electrode terminal **18** is provided on the other of the pair of short side surfaces **13**, **14**. In the example shown in FIG. **2**, the positive electrode terminal **17** is provided on the short side surface **14**, and the negative electrode terminal **18** is provided on the short side surface **13**. Note that both the positive electrode terminal **17** and the negative electrode terminal **18** may be provided on one of the pair of short side surfaces **13**, **14**.

[0030] Referring to FIG. **1**, the housing case **20** includes an upper case **21** and a lower case **22**. The lower case **22** includes a bottom plate and a peripheral wall. The bottom plate is formed in a flat plate shape. The peripheral wall is formed so as to extend upward from the outer peripheral edge portion of the bottom plate, and is formed in a ring shape. The electric power storage units **10** are housed in a space formed by assembling the upper case **21** to the lower case **22**. FIG. **1** shows the electric power storage device **1** in a state where the upper case **21** is detached. The upper case **21** corresponds to an example of a “second case” in the present disclosure, and the lower case **22** corresponds to an example of a “first case” in the present disclosure.

[0031] The electric power storage units **10** are arranged and laminated in an X direction in a space defined by the upper case **21** and the lower case **22**, whereby the electric power storage units **10** are housed in the housing case **20**. In the present embodiment, the X direction corresponds to a “lamination direction” of the present disclosure.

[0032] FIG. **3A** and FIG. **3B** are sectional views taken along III-III in FIG. **1**. FIG. **3A** shows an example in which a central portion in a longitudinal direction of the electric power storage unit **10** is bonded, and FIG. **3B** shows an example in which the entire electric power storage unit **10** is bonded.

[0033] Referring to FIG. **3A**, the bottom surface **12** of the electric power storage unit **10** and the lower case **22** are bonded to each other with an adhesive member **30** at a substantially central portion in the longitudinal direction of the electric power storage unit **10**. A protruding portion P extending in the X direction (lamination direction) is formed in the upper case **21**. The protruding portion P protrudes toward the lower case **22**. The protruding portion P abuts against the upper surface **11** of the electric power storage unit **10** at a substantially central portion in the longitudinal direction of the electric power storage unit **10**. The adhesive member **30** and the protruding portion P are located at positions where they overlap each other in a Z direction.

[0034] Referring to FIG. **3B**, the lower surface **12** of the electric power storage unit **10** and the lower case **22** are bonded to each other with an adhesive member **30a**. The adhesive member **30a** bonds substantially the entire lower surface **12** of the electric power storage unit **10** to the lower case **22**. A protruding portion P extending in the X direction (lamination direction) is formed in the

upper case **21**. The protruding portion P protrudes toward the lower case **22**. The protruding portion P abuts against the upper surface **11** of the electric power storage unit **10** at a substantially center portion in the longitudinal direction of the electric power storage unit **10**.

[0035] According to the configuration of FIG. 3A, FIG. 3B, the protruding portion P of the upper case **21** abuts against the upper surface **11** of the electric power storage unit **10**, whereby the electric power storage unit **10** receives a load on the adhesive member **30**, **30a** side (lower case **22** side) and is pressed down. This makes it possible to restrain the adhesion between the electric power storage unit **10** and the housing case **20** (lower case **22**) from being released. Furthermore, in the configuration of FIG. 3A, the adhesive member **30** and the protruding portion P are located at positions where they overlap each other in the Z direction, so that even when the central portion in the longitudinal direction of the electric power storage unit **10** is bonded to the lower case **22**, it is possible to preferably restrain the adhesion between the electric power storage unit **10** and the housing case **20** (lower case **22**) from being released.

[0036] In the configuration of FIG. 3A and FIG. 3B, a cooler for cooling the electric power storage unit **10** may be disposed in a space between the upper case **21** and the upper surface **11** of the electric power storage unit **10**. In the configuration of FIG. 3A, coolers may be disposed between the bottom surface **12** of the electric power storage unit **10** and the lower case **22** on both sides in the Y direction of the adhesive member **30**.

Embodiment 2

[0037] FIG. 4 is an exploded perspective view of an electric power storage unit **10A** according to an embodiment 2. In the embodiment 2, the electric power storage unit **10A** includes a cell assembly **50** in which a plurality of electric power storage cells **100** is electrically connected at connection portions **110**. The electric power storage cell **100** is, for example, a lithium ion battery. The electric power storage cell **100** includes, for example, an electrode body made of a wound body in which a positive electrode sheet coated with a positive electrode active material and a negative electrode sheet coated with a negative electrode active material are wound with a separator interposed therebetween, and a laminate exterior body **160** that seals the electrode body.

[0038] The electric power storage cell **100** has current collecting terminals **140** (one is a positive electrode and the other is a negative electrode) at both ends in the Y direction, and the current collecting terminals **140** of adjacent electric power storage cells **100** are electrically connected to each other in series at the connection portions **110** to form the cell assembly **50**. The cell assembly **50** is inserted into the cell case **300**, and a cover member **310** is joined to the cell case **300**, whereby the cell assembly **50** is housed in the cell case **300** to form the electric power storage unit **10A**.

FIG. 4 is a perspective view of the cell assembly **50** when the cell assembly **50** is inserted into the cell case **300**. The electric power storage unit **10A** includes the cell assembly **50** in which a plurality of electric power storage cells **100** arranged in the Y direction (connection direction) is electrically connected, and the cell case **300** that houses the cell assembly **50**.

[0039] A pair of reinforcing members **200** are provided at the connection portion **110** of the cell assembly **50** so as to interpose the current collecting terminals **140** therebetween. The reinforcing member **200** has a rectangular prism shape that is hollow in the Z direction. The material of the reinforcing member **200** may be synthetic resin or metal. The length of the reinforcing member **200** in the Z direction is the same as the width of the inner surface of the cell case **300** in the Z direction. Therefore, the reinforcing member **200** functions as a reinforcing member (so-called support rod) for the cell case **300** at the connection portion **110** of the cell assembly **50**.

[0040] Output terminals **400** (one is a positive terminal, the other is a negative terminal) are connected to the current collecting terminals **140** on both sides of the cell assembly **50**.

Furthermore, a pair of reinforcing members **210** similar to the reinforcing members **200** may be provided so as to interpose the current collecting terminals **140** on both sides of the cell assembly **50** therebetween.

[0041] FIG. 5 is a sectional view of the electric power storage device **1A** according to the

embodiment 2. This sectional view is a sectional view of a portion similar to that of FIG. 3A, FIG. 3B. In the embodiment 2, the cell assembly **50** is formed of three electric power storage cells **100**. A housing case **20A** includes an upper case **21A** and a lower case **22A**. The plurality of electric power storage units **10A** is arranged in the X direction and laminated in a space defined by the upper case **21A** and the lower case **22A**, whereby the electric power storage units **10A** are housed in the housing case **20A**.

[0042] The lower surface **12A** of the electric power storage unit **10A** (cell case **300**) and the lower case **22A** are bonded to each other with adhesive members **30A**, **30A**. The adhesive members **30A** are provided at positions where the reinforcing members **200** are arranged (the positions of the connection portions **110** of the cell assembly **50**), and bond the electric power storage unit **10A** and the lower case **22A** to each other. Protruding portions **P1**, **P2** extending in the X direction (lamination direction) are formed in the upper case **21**. The protruding portions **P1**, **P2** protrude toward the lower case **22A**. The protruding portions **P1**, **P2** abut against the upper surface **11A** of the electric power storage unit **10A** (cell case **300**). The adhesive members **30A**, **30A** and the protruding portions **P1**, **P2** are arranged at positions where they overlap each other in the Z direction.

[0043] According to the embodiment 2, the protruding portions **P1** and **P2** of the upper case **21A** abut against the upper surface **11A** of the electric power storage unit **10A** (cell case **300**), whereby the electric power storage unit **10A** receives a load and is pressed against the adhesive member **30A** (lower case **22A** side). This makes it possible to restrain the adhesion between the electric power storage unit **10A** and the housing case **20A** (lower case **22A**) from being released. Furthermore, the protruding portions **P1**, **P2** and the adhesive members **30A**, **30A** are arranged at positions where they overlap each other in the Z direction at the positions where the reinforcing members **200** are arranged. As a result, the force applied to the electric power storage unit **10A** (cell case **300**) by the protruding portions **P1**, **P2** is efficiently transmitted to the adhesive members **30A**, **30A** (adhesive points) through the reinforcing members **200**, so that it is possible to more preferably restrain the adhesion between the electric power storage unit **10A** and the housing case **20A** (lower case **22A**) from being released.

[0044] In the embodiment 2, the cell assembly **50** is configured by three electric power storage cells **100**. However, the number of electric power storage cells **100** may be two, or four or more.

Modification

[0045] FIG. **6** is a perspective view that schematically shows an electric power storage device **1B** according to a modification. In the electric power storage device **1B** of the modification, a plurality of partition walls **61**, **62**, **63** is formed in the lower case **22B** of the storage case **20B**. The partition walls **61**, **62** are formed to extend in the X direction, and the partition wall **63** is formed at the center portion in the X-direction of the lower case **22B** so as to extend in the Y direction. The plurality of electric power storage units **10** has the same configuration as in the embodiment 1. The electric power storage units **10** are arranged and laminated in the Y direction between the partition wall **61** and the partition wall **62**, whereby the electric power storage units **10** are housed in the housing case **20B**. The electric power storage units **10** are also partitioned by the partition wall **63**, whereby the electric power storage units **10** are arranged in two rows.

[0046] Each electric power storage unit **10** is bonded to the bottom surface of the lower case **22B** with an adhesive member. Protruding portions **P3**, **P4** extending in the Y direction (lamination direction) are formed in the upper case **22B**. The protruding portions **P3**, **P4** protrude toward the lower case **22B**. The protruding portions **P3**, **P4** abut against the electric power storage unit **10** at a substantially central portion in the longitudinal direction of the electric power storage unit **10**.

[0047] In this modification, the protruding portions **P3**, **P4** of the upper case **21B** abut against the electric power storage unit **10**, so that the electric power storage unit **10** receives a load on the adhesive member side (lower case **22B** side), and is pressed down. This makes it possible to restrain the adhesion between the electric power storage unit **10** and the housing case **20B** (lower

case 22B) from being released.

[0048] In the above embodiment, some of the electric power storage units (for example, the electric power storage units at the end portion in the lamination direction) do not necessarily have to abut against the protruding portions formed on the upper case.

[0049] The embodiments disclosed herein should be considered to be illustrative and not restrictive in all respects. The scope of the present disclosure is not defined by the foregoing description, but defined by the claims, and is intended to include meanings equivalent to the claims and all alterations within the claims.

Claims

1. An electric power storage device comprising: a plurality of electric power storage units; and a housing case that includes a first case and a second case, and houses the electric power storage units in a space defined by the first case and the second case, wherein: the electric power storage units are bonded to the first case, the second case has a protruding portion protruding toward the first case, and the protruding portion abuts against the electric power storage units.
 2. The electric power storage device according to claim 1, wherein the electric power storage units are arranged in a lamination direction in the storage case, and the protruding portion extends in the lamination direction.
 3. The electric power storage device according to claim 1, wherein: each of the electric power storage units has a rectangular parallelepiped shape, the electric power storage units are bonded to the first case at center portions in a longitudinal direction of the electric power storage units, and the protruding portion abuts against the electric power storage units at the center portions in the longitudinal direction of the electric power storage units.
 4. The electric power storage device according to claim 1, wherein each of the electric power storage units comprises: a cell assembly in which a plurality of electric power storage cells arranged in a connection direction is electrically connected at a connection portion; a cell case that houses the cell assembly; and a reinforcing member that is disposed at the connection portion and reinforces the cell case, wherein the protruding portion abuts against the cell case at a position where the reinforcing member is disposed.
 5. The electric power storage device according to claim 4, wherein the cell case is bonded to the first case at a position where the reinforcing member is disposed.
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